

Package insert:

Herpesan Gel

2% w/w Carbenoxolone Sodium

Name and Strength of Active Substance

The active substance is carbenoxolone sodium. Each 5g contains 100 mg carbenoxolone sodium.

Other Ingredients

The other ingredients are polyoxyethylene-30- cetylstearyl alcohol, polyol fatty acid ester, iso-octyl stearate, imidazolidinyl urea, sodium methylparahydroxybenzoic acid, sodium ethylparahydroxybenzoic acid, sodium propylparahydroxybenzoic acid, lemon oil and purified water.

Product Description

Herpesan Gel is a translucent, colourless gel with a lemon odour.

Pharmacodynamics/Pharmacokinetics

The active ingredient present in Herpesan Gel namely Carbenoxolone Sodium is derived from glycyrrhizic acid. Carbenoxolone Sodium is steroidal in structure and steroid-like activity has been demonstrated with regard to both anti-inflammatory and mineralocorticoid activity. Glycyrrhizic acid has significant anti-viral activity against a range of viruses including Herpes Simplex Type I, Vaccinia and Influenza Types A and B. Herpes Simplex Type I virus is irreversibly inactivated by glycyrrhizic acid.

In keeping with its steroid-like structure, Carbenoxolone Sodium has significant anti-inflammatory actions and suppresses rat paw oedema caused by formaldehyde and tuberculin. Carbenoxolone Sodium has about one third the potency of hydrocortisone in these studies. Other steroid-like activities include mineralocorticoid effects as a consequence of both direct aldosterone like activity of Carbenoxolone Sodium and its ability to potentiate endogenous aldosterone activity.

The efficacy of Carbenoxolone Sodium in ulcer healing is most likely related to increased mucous secretion with consequent cyto-protection. Peskar has demonstrated that Carbenoxolone Sodium inhibits the prostaglandin degrading enzymes 15-hydroxy-prostaglandin-dehydrogenase and delta ¹³-prostaglandin reductase.

General Pharmacodynamics

Smooth muscle

Carbenoxolone does not modify the action of a variety of spasmogens active on smooth muscle including acetylcholine, nicotine, histamine, 5-hydroxytryptamine or barium and does not influence gastrointestinal motility.

Cardiovascular system

The elevation of blood pressure produced by prolonged dosage with Carbenoxolone is exerted indirectly through sodium retention associated with mineralocorticoid activity.

Pharmacokinetics

Absorption following topical application to the oral mucous membrane

In clinical use for the treatment of gastric ulcer Carbenoxolone has been used in doses up to 300 mg/day. A 5 g tube of Herpesan Gel contains 100 mg Carbenoxolone. Topical application of Herpesan Gel is likely to involve the use of about 250 mg per application equivalent to 20 mg Carbenoxolone per day. It would be expected that absorption of Carbenoxolone following topical application would be much smaller than that following oral dosing. Rinsing the mouth with 30 ml of an aqueous solution containing 0.067% Carbenoxolone three times daily for one month did not produce detectable plasma Carbenoxolone levels.

After a single 200 mg oral dose, plasma carbenoxolone levels reach 30 µg/ml. Reference is made in the documentation provided to a study in which topical application of a carbenoxolone containing gel three times daily for one month produced a plasma carbenoxolone level of 5.3 µg/ml at the end of the treatment period. However this statement is uncorroborated by an original literature reference.

Metabolism and elimination

In rats given an oral dose of ¹⁴C Carbenoxolone labelled in the succinate moiety, a substantial proportion of the label is rapidly excreted as carbon dioxide. Since it would be expected that because of its place in intermediary metabolism, free succinate would rapidly appear as exhaled carbon dioxide, this finding indicates that in the rat, carbenoxolone is readily hydrolysed at the ester linkage to succinate and glycyrrhetic acid. Such hydrolysis could not be detected after incubation of carbenoxolone with blood or with liver homogenate but carbenoxolone could be hydrolysed by incubation with rat stomach or faecal contents.

It was therefore suggested that hydrolysis can take place within the gastrointestinal tract prior to absorption and that the gut microbial flora in this species are likely to be responsible for this metabolism. In rats, trituated glycyrrhetic acid was converted by the liver to the 30-glucuronide, the 3-O-glucuronide and to the 3-O-hydrogen sulphate and these metabolites were excreted into the bile.

By contrast, in man after oral administration of ¹⁴C succinate labelled carbenoxolone, a much smaller proportion of the

label (12-20%) appeared in exhaled carbon dioxide with the overwhelming majority (70-80%) appearing in the faeces. No appreciable hydrolysis was found in stomach contents even when these were buffered to pH 8. Only inconsequential amounts (0.2-1.0%) of label was excreted in the urine. It was established using TLC that the form present in faeces was carbenoxolone itself with only traces of glycyrrhetic acid detected. In human patients with T-tube drainage, one major metabolite, the 30-glucuronide was detected in bile with only traces of other metabolites including glycyrrhetic acid 3-sulphate. It is concluded that metabolism in man differs substantially from that in the rat with little hydrolysis of carbenoxolone itself within the gastrointestinal tract and with a single major glucuronide metabolite being excreted in bile and reconverted to carbenoxolone in the small intestine.

Indication

Treatment of Lip Sores, Mouth Ulcers and Orofacial Lesions caused by virus infections.

Recommended Dosage

Adults and Children, over 3 years of age:

Herpesan Gel should be applied thickly to lesions four times daily, preferably after meals and at bedtime. The gel should be applied at the first symptoms (tingling or burning sensation) of a lip sore or mouth ulcer. The gel should be allowed to remain on the affected area for as long as possible.

Herpesan Gel is for topical use only.

Route of Administration

For topical use.

Contraindications

Herpesan Gel is contraindicated in patients suffering from severe cardiac, renal and hepatic impairment, patients with hypertension, or those who are receiving cardiac glycosides. If bacterial infection is superimposed appropriate therapy should be given.

Warnings and Precautions

For topical use only.

Consult your doctor if you are an expectant mother or if the patient is a child under 3 years old before using this medicine.

Keep all medicines out of the reach of children.

Interactions with Other Medicaments

The product should not be used in the elderly or in patients with low serum albumin.

The use of Herpesan Gel is unlikely to result in any impairment of the ability of patients to drive or operate machinery.

Pregnancy and Lactation

Ask your doctor or pharmacist for advice before taking Herpesan Gel, if you are an expectant mother. No teratogenic effects have been reported with Carbenoxolone Sodium but it is not recommended that the gel should be used during the first trimester of pregnancy.

Adverse Effects/Undesirable Effects

The side effects of oral Carbenoxolone Sodium are well documented and outlined in Parts III and Part IV. The side effects of Herpesan Gel appear to be absolutely insignificant given the large number of patients treated in the various trials without any adverse side effects emerging whatsoever. Furthermore on pharmacological grounds, blood levels associated with Herpesan Gel are so low that significant side effects would not be anticipated. Given the number and variety of patients treated with Herpesan Gel in the context of clinical trials, the large clinical experience in routine clinical practice and the many decades of experience with Carbenoxolone Sodium used orally in the treatment of peptic ulcer disease, it is reasonable to suggest that no significant effects whatsoever can be anticipated from Herpesan Gel.

In summary, the risk benefit of Herpesan Gel in the treatment of a wide variety of patients with acute and recurrent lip and mouth ulceration suggests that the drug is efficacious in accelerating relief and ulcer healing but is not associated with any important side effects.

Storage Conditions

Keep all medicines out of reach of children.

Do not store above 30°C. Replace the cap tightly after use and store in a cool, dry place.

Dosage Forms and Packaging Available

Herpesan Gel is available in a 5g tube.

Name and Address of Manufacturer

Rowa Pharmaceuticals Ltd., Bantry, Co. Cork, Ireland.

Product Registration Holder

DCH Auriga (Malaysia) Sdn. Bhd. (719-V) (f.k.a. LF Asia (M) Sdn. Bhd.), Lot 6, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam, Selangor.

Date of revision of PI

This leaflet was last revised in June 2018.

